FORCES IN UNIFORM CIRCULAR MOTION

LEARNING GOALS

Students will:

- Be able to apply Newton's Laws in a situation where an object is moving in a circle

NEWTON'S 2ND LAW APPLIED TO UNIFORM CIRCULAR MOTION

Consider a ball of mass *m* that is tied to a string of length *r* and is being whirled at a constant speed in a horizontal circular path. You have learned that centripetal acceleration is caused by a net centripetal force directed toward the centre of the circle and that centripetal acceleration can be calculated by $a_c = \frac{v^2}{r}$. Combining this with Newton's second law gives an expression for the net centripetal force causing the acceleration:

$$\sum F_c = ma_c = m\frac{v^2}{r}$$

A force causing a centripetal acceleration acts toward the center of the circular path and causes a change in direction of the velocity vector. If that force should vanish, the object would no longer move in its circular path; instead, it would move along a straight-line path tangent to the circle.

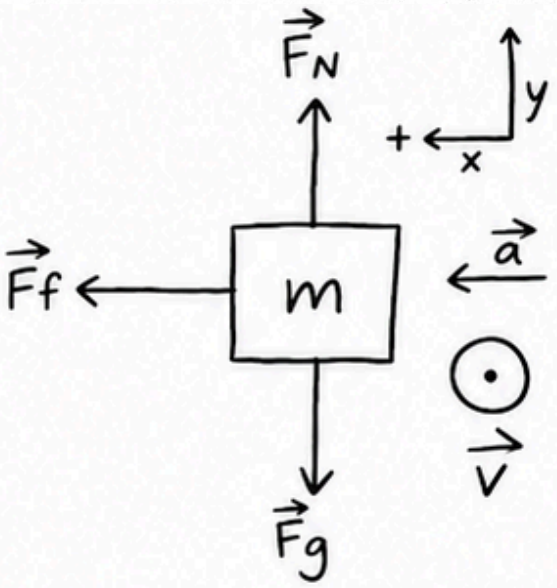
FORCE ANALYSIS IN CIRCULAR MOTION

Problems in circular motion are solved the same as any other 2nd law problems. You...

a) draw a free body diagram
b) Define the +x direction as the direction of acceleration and draw any necessary components
c) Write a scalar equation for the net force on the object (e.g. $F_{net} = T - F_g$)
d) Use Newton's second law ($\sum F_c = ma_c = m\frac{v^2}{r}$) and solve

EXAMPLE 1: CAR AROUND A FLAT CURVE

a) A car of mass m moving on a flat, horizontal road negotiates a curve. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.500, find the maximum speed the car can have and still make the turn successfully. DRAW THE FBD!



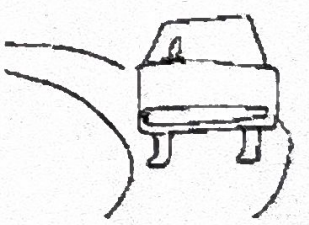
$F_f = m\frac{v^2}{r}$ $F_f = mg\mu$

$mg\mu = m\frac{v^2}{r}$

$g\mu = \frac{v^2}{r}$

$\sqrt{g\mu r} = v$

$v = 13.1\ m/s$



13 m/s

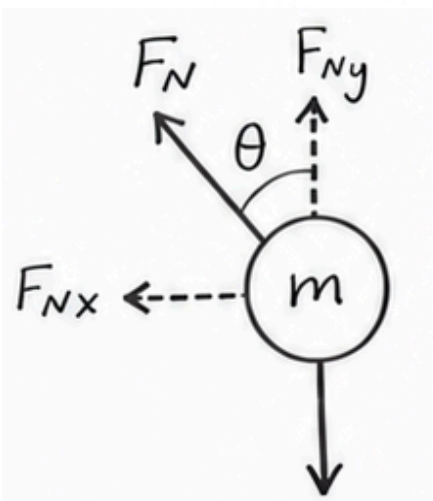
b) Based on the steps involved in this solution, why do highways not need multiple speed limits for the various masses of vehicles that travel along them?

Because mass gets canceled out and therefore is not a part of the equation to find the maximum speed.

EXAMPLE 2: BANKING THE CURVE

A civil engineer wishes to design a curved exit ramp for a highway in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a ramp is usually banked; this means the roadway is titled toward the inside of the curve.

a) Suppose the designated speed for the ramp is to be 60 km/h and the radius of the curve is 50.0 m. At what angle θ should the curve be banked? (hint: think tan)



y - dir

$F_{net,y} = F_{Ny} - F_g$

$F_N \cos\theta = mg$

$F_N = \frac{mg}{\cos\theta}$

x - dir

$F_{net,x} = F_{Nx}$

$\frac{mg}{\cos\theta} \sin\theta = F_{Nx}$

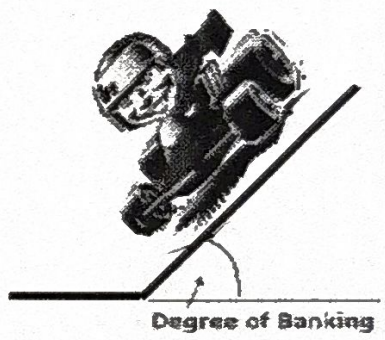
$\frac{mg}{\cos\theta} \sin\theta = m\frac{v^2}{r}$

$\frac{g}{\cos\theta} \sin\theta = \frac{v^2}{r}$

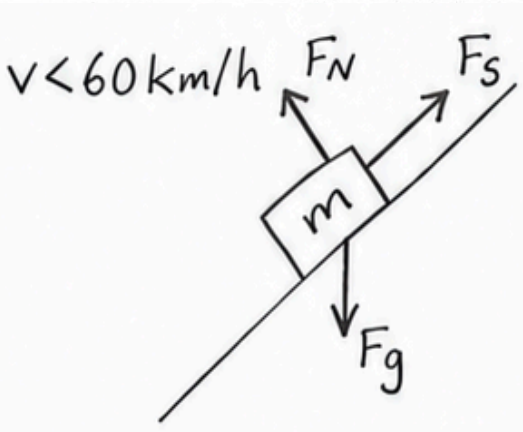
$g \tan\theta = \frac{v^2}{r}$

$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

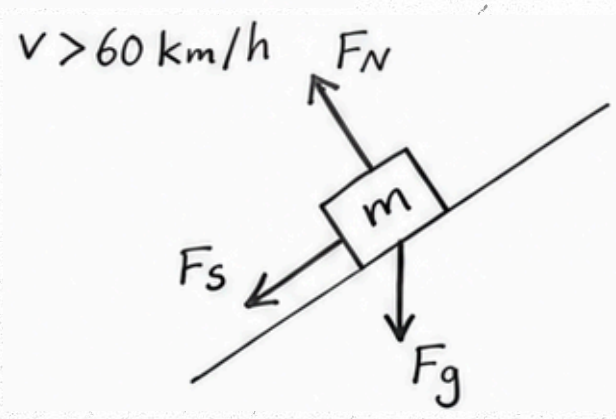
$\theta = 30°$



b) If a driver rounds the curve at a speed less than 60 km/h, which ~~force (type and~~ direction) would they need to rely on to keep them on the road? Draw an FBD showing this situation.



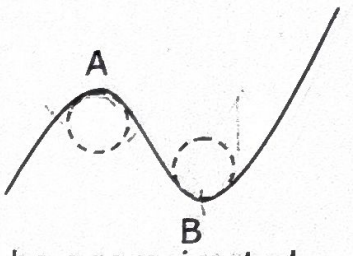
The driver would need to rely on the static friction up the bank (parallel to the road) to keep the car on the road.



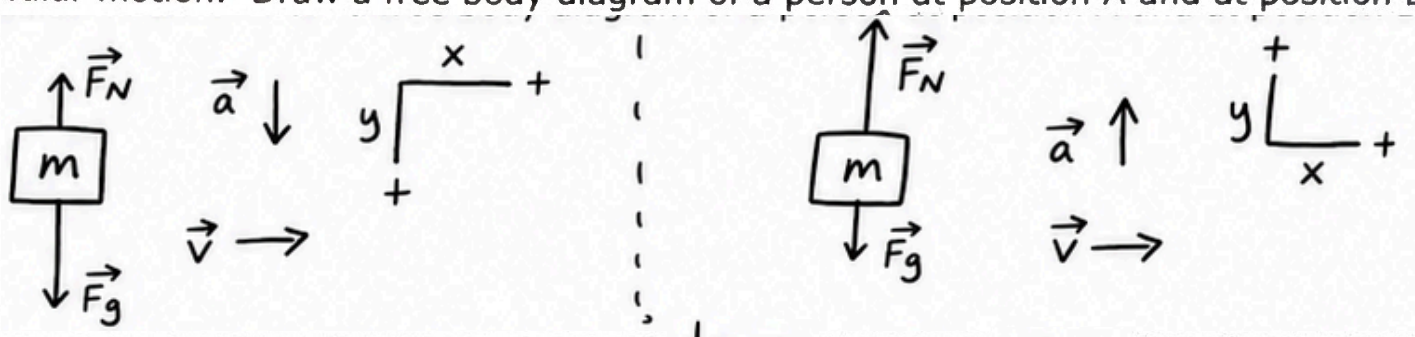
a) 30°	0-2

EXAMPLE 3: ROLLER COASTER & APPARENT WEIGHT

1. Imagine you are on a roller coaster. Do you feel heavier than usual or lighter than usual...
   a. as you go over a hill (position A)? lighter
   b. when you are at the bottom of a hill (position B)? heavier



2. As you can see in the diagram, the top and bottom portions of hills can be approximated as circular motion. Draw a free body diagram of a person at position A and at position B.



3. What **direction** is the acceleration at A? down at B? up

4. Based on your answer for 3, should the normal force be bigger or smaller than the force of gravity at A? What about at B? Fix your free body diagrams if necessary.

   smaller at A, bigger at B

Recall that our **apparent weight** is how heavy we feel. It is our perception of our weight and is calculated by the normal force on our body. We can calculate our apparent mass (m_{app}) according to

apparent weight = $N = m_{app}g$. Just realize that the mass doesn't actually change.

5. Say a person at the bottom of a hill on a roller coaster feels four times heavier than usual. Determine the speed of the roller coaster if the hill has a radius of curvature of 13.6 m.

$F_c = F_N - F_g \qquad F_N = 4F_g$

$$\frac{mv^2}{r} = 4F_g - F_g$$

$$\frac{mv^2}{r} = 3mg$$

$$\frac{v^2}{r} = 3g$$

$$v = \sqrt{3gr}$$

$$v = \sqrt{3 \times 9.81 \times 13.6)}$$

$$v = 20.0\ m/s$$

20.0 m/s

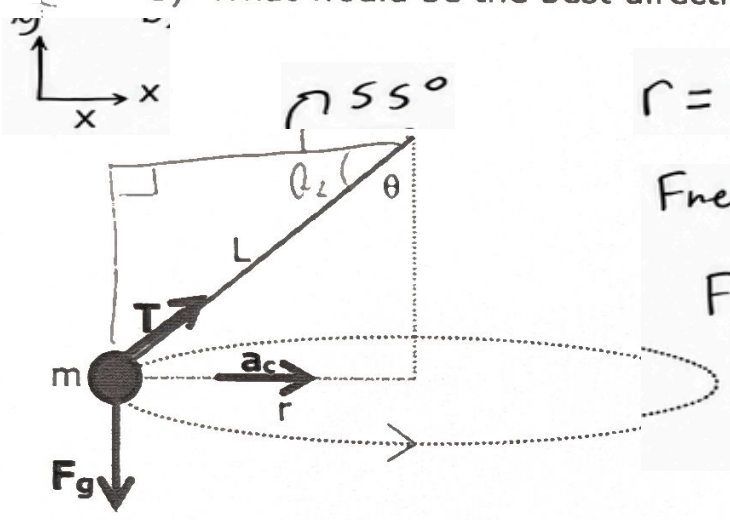
EXAMPLE 4: SWINGING A MASS IN A HORIZONTAL CIRCLE

A 0.500 kg mass is swung on a 0.90 m long string in a horizontal circle so that the string makes an angle of 35^0 to the vertical. What is the speed of the mass around the circle?

using 55° as θ

a) Draw an FBD of the mass.

b) What would be the best direction for +x? Draw appropriate components.



$r = 0.9 \cos 55°$

$F_{net\,y} = F_{Ty} - mg$

$F_T \sin\theta = mg$

$F_T = \dfrac{mg}{\sin\theta}$

x-direction

$F_{net\,x} = F_T \cos\theta$

$F_{net\,x} = \left(\dfrac{mg}{\sin\theta}\right)\cos\theta$ → $= \cot\theta$

$\dfrac{mv^2}{r} = mg\cot\theta$

$v = \sqrt{rg\cot\theta}$

$v = \sqrt{0.9\cos 55° \times 9.81 \times \dfrac{1}{\tan 55°}}$

$v = 1.9\ m/s$

VERTICAL FORCES:

c) Since the acceleration is horizontal, what happens to the vertical forces? They balance, so $F_{net\,y} = 0$

d) Use your answer above to solve for the tension force.

$F_{net\,y} = 0 \quad F_T \sin\theta - mg = 0 \quad F_T = mg/\sin\theta \quad F_T = \dfrac{mg}{\sin\theta}$

$F_T = \dfrac{0.5 \times 9.81}{\sin 55°} \quad F_T = 6.0\ N$

HORIZONTAL FORCES:

e) What is the only horizontal force? X-component of tension, $F_{Tx} = F_T \cos\theta$

f) Use Newton's second law to solve for the centripetal acceleration.

$F_{net\,x} = ma_x \quad a_x = \dfrac{F_{net\,x}}{m}$

$a_x = \left(\dfrac{mg}{\sin\theta}\right)\cos\theta \div m$

$a_x = \dfrac{0.5 \times 9.81}{\sin 55°}\cos 55° \div 0.5$ → $a_x = 6.87\ m/s^2$

$\vec{a}_c = 6.9\ m/s^2$ [towards the centre]

g) Finally, find the speed from the centripetal acceleration. Remember, you are given the length of the string, but this is NOT the radius of the circle. You have to find r first. (See the diagram above.)

$r = L\cos\theta$

$a_c = \dfrac{v^2}{r}$

$v = \sqrt{a_c r}$ → $v = \sqrt{a_c (L\cos\theta)}$

$v = \sqrt{6.9 \times 0.9\cos 55°}$

$v = 1.9\ m/s$

matches the speed calculated earlier

d) 6.0 N
f) 6.9 m/s/s
g) 1.9 m/s

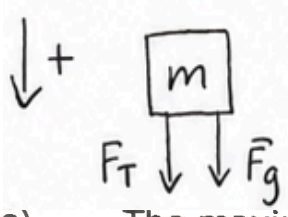
EXAMPLE 5: SWINGING A MASS IN A VERTICAL CIRCLE

A 0.500 kg mass is swung on a 65 cm long string in a vertical circle with a frequency of 1.2 Hz.

a) Calculate the centripetal acceleration of the mass.

$a_c = 4\pi^2 r f^2 \quad a_c = 4\pi^2 \times 0.65 \times 1.2^2 \quad a_c = 37\ m/s^2$

b) Find the tension in the string at the top of the circle. Draw an FBD.



$F_{net y} = ma_c$

$ma_c = F_T + F_g$

$F_T = ma_c - mg$

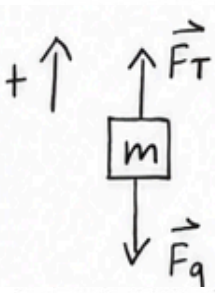
$F_T = 0.5 \times 4\pi^2 \times 0.65 \times 1.2^2 - (0.5 \times 9.81)$

$F_T = 14\ N$ → at the top

13.6 N unrounded

c) The maximum tension the string can take is 20 N. Will the string break at any point during the swing? If so, will it happen near the top or the bottom?

max tension is at the bottom



$ma_c = F_T - F_g$

$F_T = ma_c + F_g$

$F_T = 0.5 \times 4\pi^2 \times 0.65 \times 1.2^2 + (0.5 \times 9.81)$

$F_T = 23\ N$ at bottom

It will break near the bottom because the required tension is 23 N, greater than the 20 N limit.

a) 37 m/s²
b) 13.6 N
c) (T_{max} = 23 N)

EXAMPLE 5: MINIMUM SPEED

When you swing a mass on a string in a vertical circle, there is a minimum speed needed to maintain the circle. If you go too slow the mass just starts to FALL along a parabolic path inside the circular path it was following. Let's find that minimum speed.

In the problem you did above, you have only two forces. One force will not change no matter how slowly you swing the mass.

a) Which force will not change? Force of gravity

b) Which force will change? Force of tension

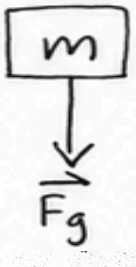
What is the minimum possible value for this force? 0 N

A 0.500 kg mass is swung on a 65 cm long string in a vertical circle at the slowest speed needed to maintain circular motion. Draw the FBD for this situation.

a) What is the net centripetal force when it is at this lowest speed?

$F_c = F_g + F_T \quad F_c = 0.5 \times 9.81 + 0 \quad F_c = 4.9\ N$

b) What is the lowest speed for the mass in this situation?



$ma_c = \frac{mv^2}{r} \quad ma_c = F_c$

$F_c = \frac{mv^2}{r}$

$v = \sqrt{(F_c \times r)/m}$

$v = \sqrt{(4.905 \times 0.65)/0.5}$

$v = 2.5\ m/s$

a) 4.9 N
b) 2.5 m/s

HOMEWORK: DO P123 Q1-5 P124 Q1-6, P142 Q32-42

PRACTICE

1. A 200. g ball on the end of a string is rotated in a horizontal circle of radius 10.0 m. The ball completes 10 rotations in 5.0 s. What is the centripetal force of the string on the ball? (316 N)
2. In the Bohr model of the hydrogen atom, the electron revolves around the nucleus. If the radius of the orbit is 5.3 x 10-11 m and the electron makes 6.6×10^{15} rotations per second, find: a. the acceleration of the electron (9.1×10^{22} m/s^2) b. the centripetal force acting on the electron – due to the electrostatic attraction between the nucleus and electron. The mass of the electron is 9.1×10^{-31} kg. (8.3×10^{-8} N)
3. A string pendulum 1.12 m long has a bob with a mass of 200 g. a. What is the tension in the string when the pendulum is at rest. (1.96 N) b. What is the tension at the bottom of the swing. If the pendulum is moving at 1.2 m/s^2? (2.2 N)
4. When you whirl a ball on a cord in a vertical circle, you find a critical speed at the top for which the tension in the cord is zero. This is because the force of gravity on the object itself supplies the necessary centripetal force. How slowly can you swing a 2.5 kg ball like this so that it will just follow a circle with a radius of 1.5 m? (3.8 m/s)
5. An object of mass 3.0 kg is whirled around in a vertical circle of radius 1.3 m with a constant velocity of 6.0 m/s. Calculate the maximum and minimum tension in the string. (1.1×10^2 N, 54 N)
6. Snoopy is flying his vintage war plane in a "loop the loop" path chasing the Red Baron. His instruments tell him the plane is level at the bottom of the loop and travelling with a speed of 180 km/h. He is sitting on a set of bathroom scales and notes that they read four times the normal force of gravity on him. What is the radius of the loop? (85 m)
7. An Australian bushman hunts kangaroos with the following weapon, a heavy rock tied to one end of a light vine of length 2.0 m. He holds the other end above his head, at a point 2.0 m above ground level and swings the rock in a horizontal circle. The cunning kangaroo has observed that the vine always breaks when the angle measured between the vine and the vertical reaches 60.0°. At what minimum distance from the hunter can the kangaroo stand with no danger of a direct hit? (2.4 m)
8. A highway has a curve of radius 180 m. Traffic is expected to move around this curve at a speed of 100. km/h. What is the banking angle required for the road surface on the curve if there is no friction? (24°)
9. A 2.0 kg stone at the end of a string 0.50 m long is whirled in a horizontal circle. If the magnitude of the maximum tension that the string can withstand is 36 N, calculate the speed of rotation which would break the string. (3.0 m/s)
10. A 45 kg girl swings on a 2.4 m long light swing with two ropes. At the lowest point in her swing, the girl's speed is 5.0 m/s. Calculate the magnitude of the tension in each rope of the swing when the girl is at the lowest point of her swing. (460. N)
11. A 0.50 m long rope is used to whirl a 1.0 kg mass in a vertical circle. When the mass was at the highest point, its speed was 2.5 m/s. Calculate the magnitude of the tension in the rope when the mass is at its highest point. (2.7 N)
12. A bridge over a small river has a roadway which is in the shape of an arch having a radius of curvature of 41 m. What is the maximum speed at which an automobile can travel across the bridge without leaving the bridge? (20. m/s)
13. A motorcycle and rider perform the stunt of circling in a cylindrical track with its axis horizontal. The diameter of the track is 10. m. The total mass of the rider and motorcycle is 200. kg. a. What is the force with which the motorcycle presses against the track at the highest point if the speed is 8.0 m/s? (6.0×10^2 N[up]) b. What is the minimum speed at the highest point if the motorcycle is not to fall from the track? (7.0 m/s)

14. A car rounds a level curve of radius 65 m at a speed of 18 m/s. What is the minimum value of the coefficient of kinetic friction between the tires and the road to prevent skidding? (0.51)

15. An auto track has semicircular ends of radius 150 m. The curves are banked at an angle of 52°. At what speed must the cars be driven to avoid any sidethrust on the wheels (i.e. so that no friction is required to maintain the turn)? (43 m/s)

16. A car having a mass of 1200 kg, rounds a curve which is banked at an angle of 18° to the horizontal. The radius of curvature is 130 m. Find the speed of the car if the frictional force between the tires and the road is 3.5×10^3 N down the banked curve. (29 m/s)

17. Patrons on an amusement park ride called the Rotor stand with their backs against the wall of a rotating cylinder while the floor drops away beneath them. To keep from sliding downward, they require a centripetal acceleration in excess of about 25 m/s^2. The Rotor has a diameter of 5.0 m. What is the minimum frequency for its rotation? How does the person not fall out? (0.50 Hz)